Module 12

Introduction to GitHub administration

Agenda

GitHub Foundations Part 1

1 Introduction to Git

2 Introduction to GitHub

3 Introduction to GitHub's products

4 Configure code scanning on GitHub

5 Introduction to GitHub Copilot

6 Code with GitHub Codespaces

7 Manage your work with GitHub Projects

8 Communicate effectively on GitHub using Markdown

GitHub Foundations Part 2

9 Contribute to an open-source project on GitHub

10 Manage an InnerSource program by using GitHub

11 Maintain a secure repository by using GitHub best practices

12 Introduction to GitHub administration

13 Authenticate and authorize user identities on GitHub

14 Manage repository changes by using pull requests on GitHub

15 Search and organize repository history by using GitHub

16 Using GitHub Copilot with Python

Introduction

In this module, you'll learn about GitHub administrative tasks at different hierarchical levels, configuring authentication for web and git client access, and the permissions associated with each hierarchical level.

By the end of this module, you'll be able to:

- Summarize the organizational structures and permission levels that GitHub administrators can use to organize members to control access and security.
- Identify the various technologies that enable a secure authentication strategy, allowing administrators to centrally manage repository access.
- Describe the technologies required to centrally manage teams and members using existing directory information services and how you can use GitHub itself as an identity provider for authentication and authorization.

What is GitHub administration?

RBAC

Entra ID

App Registration

class

Enterprise App

instance

Administration at Team Level

- **Team Creation and Management:** Create teams with cascading permissions, sync with identity providers like Microsoft Entra ID, and manage team members and outside collaborators.
- **Best Practices:** Reflect company hierarchy with nested teams, streamline PR reviews with interest-based teams, and enable team synchronization to reduce manual updates.

Administration at Organization Level

- **Organization Management:** Invite and remove members, organize users into teams, manage repository permissions, and set up security and billing.
- **Best Practices:** Set up a single organization to avoid duplication, reduce errors, and manage costs effectively.

Administration at Enterprise Level

- **Enterprise Management:** Enable SAML single sign-on, add or remove organizations, manage billing, and set up enterprise-wide policies.
- **Best Practices:** Use custom scripts for extracting information and applying changes across the enterprise.

How does GitHub authentication work?

Authentication Methods

- **Username and Password:** Basic authentication is risky for sensitive information; alternatives are recommended.
- **Personal Access Tokens (PATs):** Tokens generated via GitHub settings for API or command line authentication, tied to specific permissions.

SSH and Deploy Keys

- **SSH Keys:** Authenticate to remote servers without supplying username and token for every interaction.
- **Deploy Keys:** Grant access to a single repository, with the public key attached to the repository and the private key on the user's server.

Additional Security Options

- **Two-Factor Authentication (2FA):** Adds an extra layer of security by requiring a code from a mobile device or SMS in addition to the password.
- **SAML Single Sign-On (SSO):** Allows centralized management of user identities and secure access to organization resources via an IdP.

Best Practices

- **Team Synchronization:** Sync GitHub teams with identity providers to automate updates and reduce tasks.
- **Security Policies:** Enforce security policies at the organization and enterprise levels to protect resources.

How do GitHub organization and permissions work?

pub priv

You can customize access to a given repository by assigning permissions. There are five repository-level permissions:

- **Read:** Recommended for non-code contributors who want to view or discuss your project. This level is good for anyone that needs to view the content within the repository but doesn't need to actually make contributions or changes.
- **Triage:** Recommended for contributors who need to proactively manage issues and pull requests without write access. This level could be good for some project managers who manage tracking issues but don't make any changes.
- **Write:** Recommended for contributors who actively push to your project. Write is the standard permission for most developers.
- **Maintain:** Recommended for project managers who need to manage the repository without access to sensitive or destructive actions.
- **Admin:** Recommended for people who need full access to the project, including sensitive and destructive actions like managing security or deleting a repository. These people are repository owners and administrators.

How do GitHub organization and permissions work? (cont.)

Organization permission levels

There are six levels of permissions at the organizational level.

Permission level	Description
Owner	Organization owners can do everything that organization members can do, and they can add or remove other users to and from the organization. This role should be limited to no less than two people in your organization.
Member	Organization members can create and manage organization repositories and teams.
Moderator	Organization moderators can block and unblock nonmember contributors, set interaction limits, and hide comments in public repositories that the organization owns.
Billing manager	Organization billing managers can view and edit billing information.
Security managers	Organization security managers can manage security alerts and settings across your organization. They can also read permissions for all repositories in the organization.
Outside collaborator	Outside collaborators, such as a consultant or temporary employee, can access one or more organization repositories. They aren't explicit members of the organization.

Summary

In this module, you learned about GitHub administrative tasks at different hierarchical levels, configuring authentication for web and git client access, and the permissions associated with each hierarchical level.

You learned how to:

- Summarize the organizational structures and permission levels that GitHub administrators can use to organize members to control access and security.
- Identify the various technologies that enable a secure authentication strategy, allowing administrators to centrally manage repository access.
- Describe the technologies required to centrally manage teams and members using existing directory information services and how you can use GitHub itself as an identity provider for authentication and authorization.

End of presentation